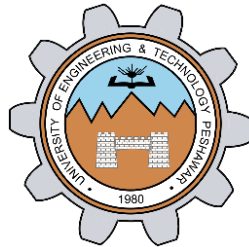


LAB # 03
INTRODUCTION TO XILINX ISE AND Spartan 6 BOARD



Spring 2024
CSE-308L Digital System Design Lab

Submitted by: **Shahzad Bangash**

Registration No: **21PWCSE1980**

Class Section: **C**

“On my honor, as student at University of Engineering and Technology, I have neither given nor received unauthorized assistance on this academic work.”

Submitted to:

Engr. Shahzada Fahim Jan

March 6, 2024

Department of Computer Systems Engineering
University of Engineering and Technology, Peshawar

Objectives of The Lab:

- Introduction to FPGA
- Introduction to Xilinx ISE

Software Used:

- Xilinx ISE

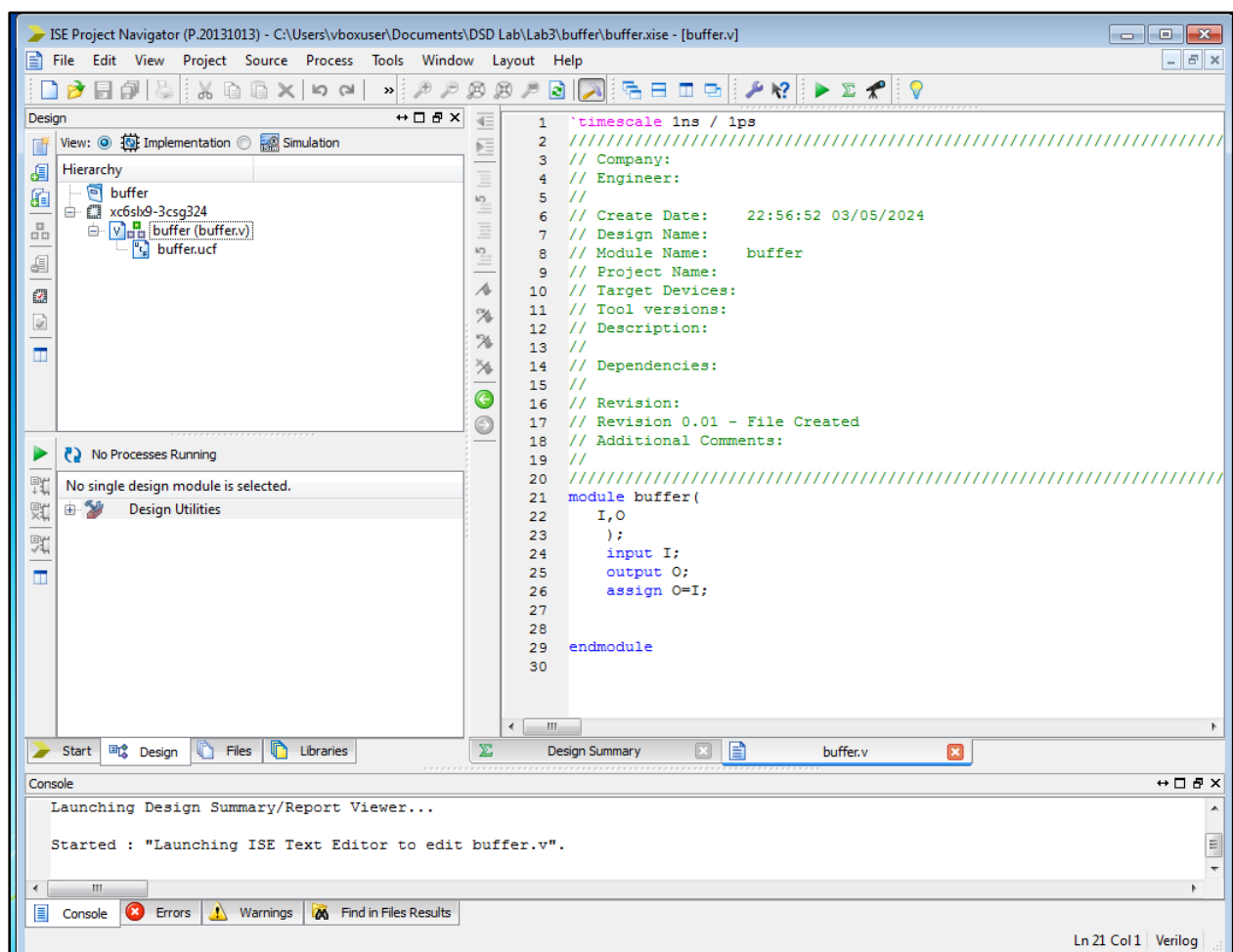
Lab Tasks:

1. Implement buffer ON the Kit and attach the snapshot.
2. Implement AND/OR/XOR gate on the Kit and attach the snapshot.
3. Implement OR gate using NAND gate.

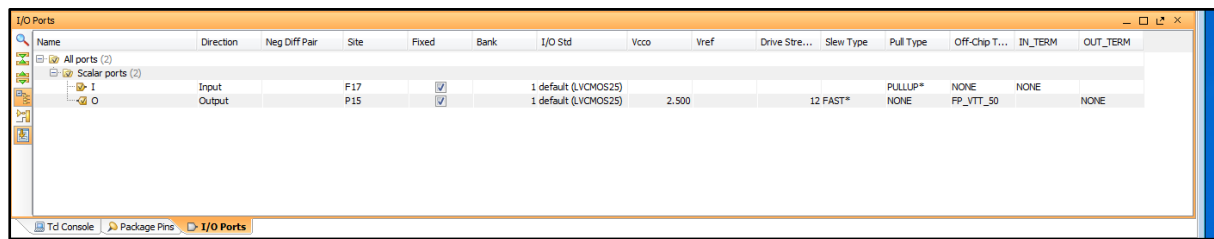
Task 1:

Implement buffer ON the Kit and attach the snapshot.

Xilinx ISE Software:



PlanAhead-Post Synthesis:



Name	Direction	Neg Diff Pair	Site	Fixed	Bank	I/O Std	Vcco	Vref	Drive Stre...	Slew Type	Pull Type	Off-Chip T...	IN_TERM	OUT_TERM
I	Input		F17	☒		1 default (LVCMOS25)					PULLUP*	NONE	NONE	
O	Output		P15	☒		1 default (LVCMOS25)	2.500		12 FAST*		NONE	FP_VTT_50	NONE	NONE

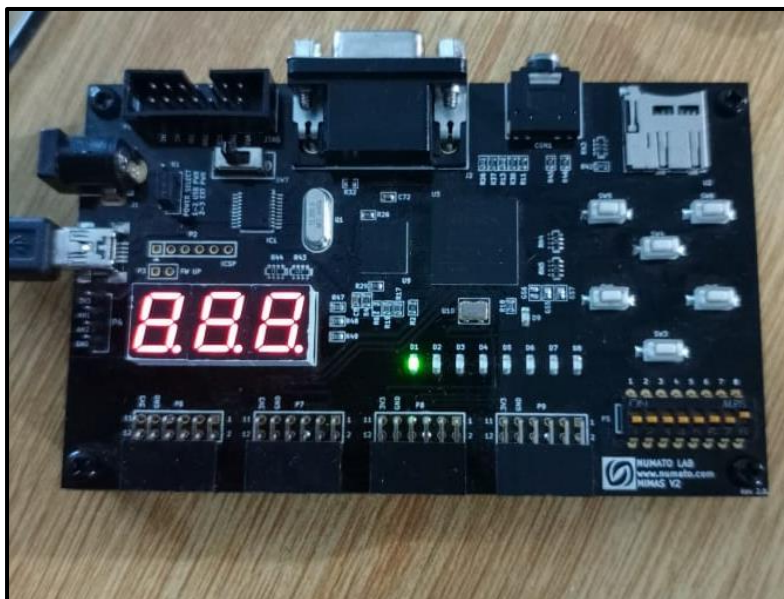
Verilog Code:

```
20
21 module buffer(
22     I,O
23 );
24     input I;
25     output O;
26     assign O=I;
27
28
29 endmodule
```

User constraints File (.ucf):

```
1
2 # PlanAhead Generated physical constraints
3
4 NET "I" LOC = F17;
5 NET "O" LOC = P15;
6
7 # PlanAhead Generated IO constraints
8
9 NET "O" SLEW = FAST;
10 NET "I" PULLUP;
11
```

Practical Implementation:



Task 2:

Implement AND gate on the Kit and attach the snapshot.

AND Gate:

Verilog Code:

```
21 module AndGate(  
22     A,B,O  
23 );  
24     input A,B;  
25     output O;  
26     and a1(O,A,B);  
27  
28 endmodule
```

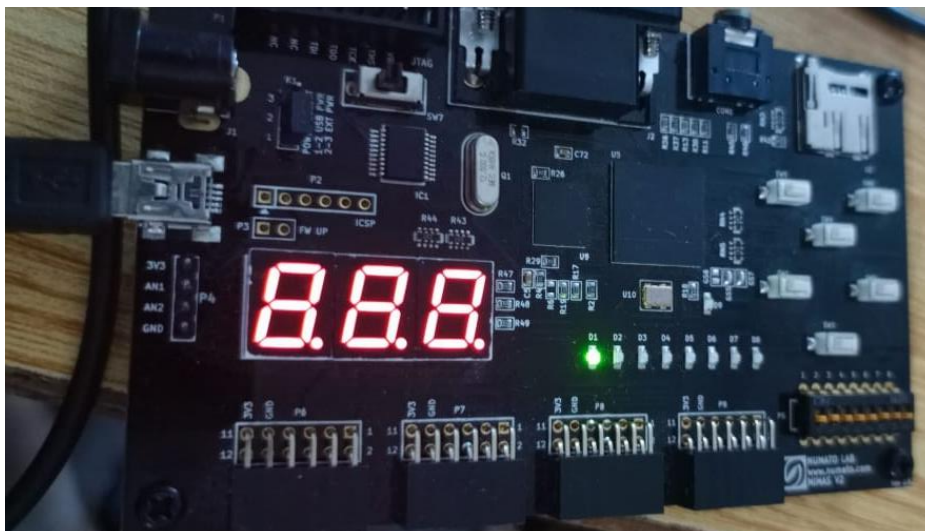
PlanAhead-Post Synthesis:

Name	Direction	Neg Diff Pair	Site	Fixed	Bank	I/O Std	Vcco	Vref	Drive Stre...	Slew Type	Pull Type	Off-Chip T...	IN_TERM	OUT_TERM
All ports (3)														
Scalar ports (3)														
A	Input		F17	☒		1 default (LVCMOS25)					PULLUP*	NONE	NONE	
B	Input		F18	☒		1 default (LVCMOS25)					PULLUP*	NONE	NONE	
O	Output		P15	☒		1 default (LVCMOS25)	2.500			12 FAST*	NONE	FP_VTT_50		NONE

User constraints File (.ucf):

```
2 # PlanAhead Generated physical constraints  
3  
4 NET "A" LOC = F17;  
5 NET "B" LOC = F18;  
6 NET "O" LOC = P15;  
7  
8 # PlanAhead Generated IO constraints  
9  
10 NET "O" SLEW = FAST;  
11 NET "A" PULLUP;  
12 NET "B" PULLUP;  
13
```

Practical Implementation:



OR Gate:

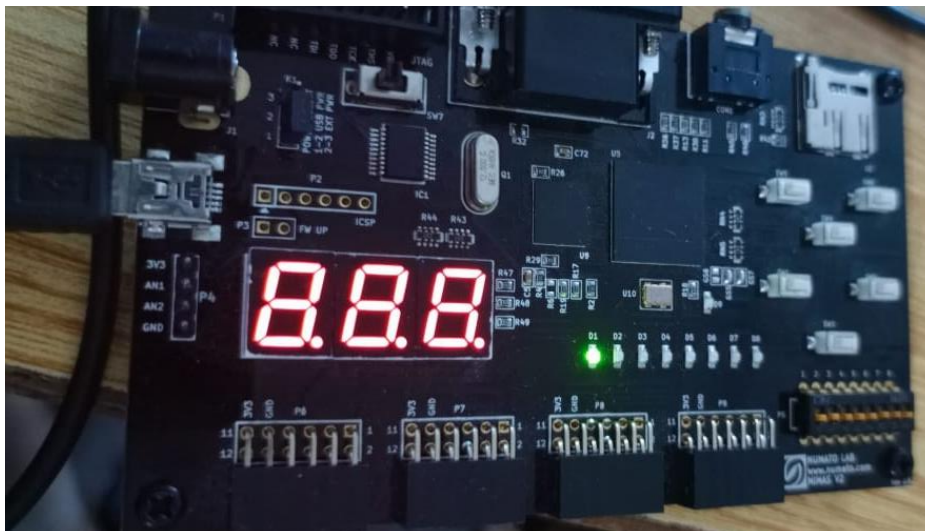
Verilog Code:

```
21 module ORGate(  
22     A,B,O  
23 );  
24     input A,B;  
25     output O;  
26     or o1(O,A,B);  
27  
28 endmodule  
29
```

User constraints File (.ucf):

```
2 # PlanAhead Generated physical constraints  
3  
4 NET "A" LOC = F17;  
5 NET "B" LOC = F18;  
6 NET "O" LOC = P15;  
7  
8 # PlanAhead Generated IO constraints  
9  
10 NET "O" SLEW = FAST;  
11 NET "A" PULLUP;  
12 NET "B" PULLUP;  
13
```

Practical Implementation:



XOR Gate:

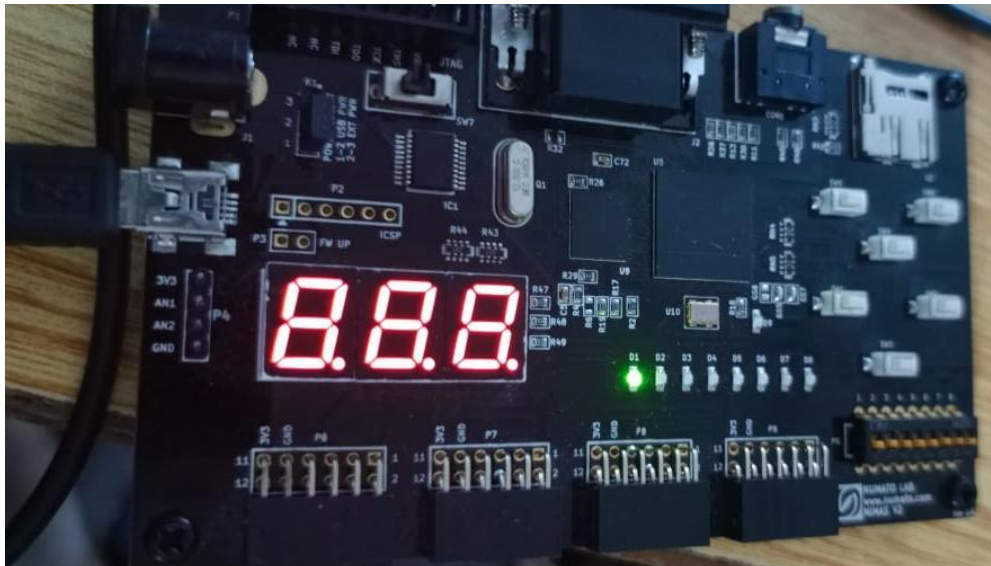
Verilog Code:

```
21 module XORGate(  
22     A,B,O  
23 );  
24     input A,B;  
25     output O;  
26     xor x1(O,A,B);  
27  
28 endmodule  
29
```

User constraints File (.ucf):

```
2 # PlanAhead Generated physical constraints  
3  
4 NET "A" LOC = F17;  
5 NET "B" LOC = F18;  
6 NET "O" LOC = P15;  
7  
8 # PlanAhead Generated IO constraints  
9  
10 NET "O" SLEW = FAST;  
11 NET "A" PULLUP;  
12 NET "B" PULLUP;  
13
```

Practical Implementation:



Task 3:

Implement OR gate using NAND gate.

Verilog Code:

```
23 module OrGate(O,A,B);
24     input A,B;
25     output O;
26
27     wire nA,nB;
28
29     nand n1(nA,A);
30     nand n2(nB,B);
31     nand n3(O,nA,nB);
32 endmodule
```

PlanAhead-Post Synthesis:

Name	Direction	Neg Diff Pair	Site	Fixed	Bank	I/O Std	Vcco	Vref	Drive Stre...	Slew Type	Pull Type	Off-Chip T...	IN_TERM	OUT_TERM
All ports (3)														
Scalar ports (3)														
A	Input		F17	☒		1 default (LVCMOS25)					PULLUP*	NONE	NONE	
B	Input		F18	☒		1 default (LVCMOS25)					PULLUP*	NONE	NONE	
O	Output		P15	☒		1 default (LVCMOS25)	2.500		12 FAST*		NONE	FP_VTT_50		NONE

User constraints File (.ucf):

```
2 # PlanAhead Generated physical constraints
3
4 NET "A" LOC = F17;
5 NET "B" LOC = F18;
6 NET "O" LOC = P15;
7
8 # PlanAhead Generated IO constraints
9
10 NET "O" SLEW = FAST;
11 NET "A" PULLUP;
12 NET "B" PULLUP;
13
```

Practical Implementation:

